Jiya Singhal

CS Undergrad — AI/ML Engineering & Product Development

Portfolio

+91-8630588622 jiya.23bcs10043@ms.sst.scaler.com linkedin.com/in/jiya-singhal github.com/jiya-singhal

Education

Scaler School of Technology, Bangalore **Expected May 2027**

*B.Sc. (Hons.) in Computer Science — **Birla Institute of Technology and Science***

Experience

SingOneSong **Bangalore, India**

Software Engineering Intern *Aug 2025 – Present*

- **Real-Time Noise Detection System:** Designed and built a multi-metric audio quality classifier from scratch using SNR, spectral entropy, A-weighted dB, and voice activity detection to separate environmental noise from loud singing in real time. Validated across **355 production recordings**, iterating through 70/30 weighted scoring with tiered dBA thresholds to **reduce false positive warnings from 5.5% to 0.6%**.
- **Pitch Detection Benchmarking (37K+ Tests):** Architected an end-to-end testing framework using Playwright to evaluate pitch detection accuracy under real-world noise. Tested **30 VocalSet singers × 150 MUSAN noise files × 5 noise levels**, comparing PESTO vs Aubio algorithms. PESTO achieved **96.8% accuracy vs Aubio's 31.3%** (Aubio collapsed to 1.4% on male voices). Identified optimal confidence thresholds and range estimator settings that shipped to production and were later ported to Unity WebGL games.
- **Onboarding API Parallelization:** Re-architected the voice analysis onboarding endpoint from sequential to concurrent execution using `asyncio.gather()`, orchestrating pitch analysis, vocal separation, gender detection, and artist enrichment as parallel tasks on GCP Cloud Run. **Reduced end-to-end response time from 57s to under 15s** (74% reduction). Also integrated a Silero VAD preprocessing step to trim silence before ML inference.
- **On-Device Model Migration (97.5% Size Reduction):** Replaced a 16MB PESTO ONNX pitch detection model with a **389KB swift-f0 model** on web, implementing the full Web Worker → ONNX Runtime WASM → AudioWorklet pipeline with 16kHz resampling and hop-256 inference. Created a preloader abstraction for near-instant model initialization, validated against VocalSet ground truth. Also replaced the onboarding gender classifier (**80% → 95% accuracy**) with a Resemblyzer-based model.
- **Cross-Platform Audio Pipeline Fixes:** Ported evidence-backed pitch detection fixes to Unity WebGL games — raised confidence thresholds, added median filtering, tuned adaptive noise floors, and applied browser audio constraint fixes (`getUserMedia` with disabled `echoCancellation/noiseSuppression`). Also diagnosed and fixed a **50ms audio desync** in the Flutter mobile app caused by MP3 encoder delay, implementing per-source latency compensation.

TradeIndia **Noida, India**

Product & Tech Intern (Voice AI, Fraud Detection) *Jan 2025 – April 2025*

- **Smart Product Search Engine:** Built a hybrid search system combining **FAISS vector search** with sentence embeddings and **FuzzyWuzzy** string matching, scoring results through a weighted blend of semantic similarity, fuzzy match, and heuristic rules. Deployed via Streamlit across thousands of product listings, **improving search relevance from 89% to 96% accuracy**.
- **Automated Voice Bot PoC:** Led end-to-end proof-of-concept testing for a customer inquiry call bot, analyzing **thousands of call transcripts** to iteratively refine scripts and improve speech recognition accuracy. Collaborated directly with vendors on tuning delivery, tone, and response handling to **reduce manual call workload** and increase customer response speed.
- **Fraud Detection Integration:** Integrated third-party verification tools (**Truecaller, IDfy, Shield**) into the platform through vendor evaluation and API coordination, strengthening trust and safety for buyers and sellers.

Technical Skills

Languages: Python, JavaScript/TypeScript, Java, SQL, HTML/CSS

ML & Audio: ONNX Runtime, Librosa, FAISS, Resemblyzer, PyTorch, TensorFlow/Keras, Signal Processing

Models & Techniques: Pitch Detection (PESTO, Aubio, swift-f0), Whisper, CREPE, Silero VAD, GRU/LSTM, Sentence Embeddings

Frameworks: FastAPI, React, Streamlit, Flutter

Cloud & Tools: GCP (Cloud Run, Firestore), Firebase, Docker, Git, GitHub Actions CI/CD, Playwright, asyncio

Highlights & Certifications

- Certified Bharatanatyam dancer and currently training in vocals.
- Targeting AI/ML engineering and product roles — passionate about building intelligent systems that ship to real users, from model selection and evaluation to production deployment.
- Product Analytics Micro-Certification — Product School, 2025. Certificate